

Geeta Chandra Raju, Bethala

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EDUCATION

New York University

New York, USA

Masters in Mechatronics and Robotics (GPA: 3.867)

September 2022 - May 2024

Coursework: Robot Perception, Deep-Learning, Advanced Mechatronics, Swarm Robotics, Sim Tools for Robotics.

Thesis: Adapt: Adapting Human Behavior in Social Navigation and Task Learning.

Acharya Nagarjuna University College of Engineering and Technology

Guntur, India

Bachelor of Technology in Mechanical Engineering

August 2017 - August 2021

Coursework: Machine Design, Material Science, Mechatronics, Automatic Control Engineering, Manufacturing.

SKILLS

Languages & Tools: C, Python, MATLAB, ROS, ROS2, Isaac Lab, Isaac Sim, Robot Web Tools, Unity 3D (AR VR), Gazebo, Arduino, Simulink, ReactJS, CATIA, SIMULIA, DELMIA, Fusion360, Solidworks, ABACUS

PUBLICATIONS, PATENTS AND MAJOR CONTRIBUTIONS

Accepted

- **H2-COMPACT: Human-Humanoid Co-Manipulation via Adaptive Contact Trajectory Policies** - IEEE International Conference on Humanoid Robots (Humanoids) 2025 [Project-page](#).
- **Wavelet Policy: Lifting Scheme for Policy Learning in Long-Horizon Tasks** - IEEE/CVF International Conference on Computer Vision (ICCV) 2025 [Paper](#).
- **Socially-Aware Robot Navigation Enhanced by Bidirectional Natural Language Conversations Using Large Language Models** - IEEE International Conference on Intelligent Robots and Systems (IROS) 2025 [Project-page](#).
- **Immersive Social Interaction with VR and LLM-Assisted Humanoids** - IEEE Humanoids 2024 workshop spotlight.
- **GAMap: Zero-Shot Object Goal Navigation with Multi-Scale Geometric-Affordance Guidance** - Conference on Neural Information Processing Systems (NeurIPS) 2024 [Project-page](#).

Under-Review

- **Humanoid Agent via Embodied Chain-of-Action Reasoning with Multimodal Foundation Models for Zero-Shot Loco-Manipulation** - [Project-page](#).
- **Hierarchical Scoring with 3D Gaussian Splatting for Instance Image-Goal Navigation** - [Project-page](#)
- **Humanoid Bimanual Dexterous Manipulation Driven by Egocentric Video** - [Project-page](#)

Patents

- **Brain-Controlled Wheelchair System for Disabled Children** - South African Patent, Official Application No: 2022/07203.
- **IoT-Based Cloud-Engaged Herbal Liquid Carrier for Logistics** - German Patent, DNR: 202022103189.4, Code: G11586DE.

EXPERIENCE

New York University - Abu Dhabi (Research Engineer)

August 2024 - Present

- Conduct research in pHRI and embodied intelligence, developing intention-aware, force-adaptive algorithms for fluid human-robot and robot-environment interaction in contact-rich tasks.
- Design and execute robotics experiments across humanoid and mobile platforms, building experimental setups, sensing and data-collection pipelines, and ROS/ROS2-based control frameworks.
- Customize, calibrate, and troubleshoot humanoids and mobile manipulators, improving hardware, communication, and control performance.

- Collaborate with faculty, postdocs, and industry engineers (Unitree, TiTa) to enhance robot functionality, resolve integration issues, and support lab-wide robotics initiatives.

Embodied AI and Robotics Lab - NYU (Research Assistant)

September 2023 - May 2024

- Developed socially aware navigation systems in ROS by integrating LLMs and vision-language models to improve human-robot interaction.
- Researched commonsense reasoning and spatial perception using LLMs and CLIP, enabling robots to better interpret social and environmental cues.
- Built and evaluated algorithms for social cue understanding in dynamic environments, combining robotics, ML, and NLP methods.

AI4CE lab - NYU (Research Assistant)

March 2023 - August 2023

- Worked on a city-scale mapping project, designing and calibrating sensor-integrated mobile platforms with 360° LiDAR, cameras, and GPS.
- Implemented ROS-based data synchronization and sensor integration, developing modular, detachable sensor mounts for flexible field deployment.
- Performed precision sensor calibration, improving the accuracy and reliability of mapping data for autonomous vehicle and urban research applications.

The Center for K12 STEM Education - NYU (STEM Leader & Trainer) Teaching

June 2023 - August 2023

- Taught sound, microcontrollers, electronics, and basic robotics to 30 middle school and 27 high school students, and prepared instructional materials and experiments for these modules.
- Trained and supported a team of 11 instructors, enabling them to deliver effective robotics and programming lessons across diverse student groups.

Mechatronics, Controls, and Robotics Lab - NYU (Research Assistant)

December 2022 - March 2023

- Developed a ROS-based navigation stack for a mobile manipulator with dynamic obstacle avoidance, and integrated web and Unity interfaces for teleoperation of medical devices.
- Built a test rig for medical manipulators to measure speed, force, precision, and repeatability, supporting task learning and performance evaluation.

AWARDS AND HONORS

Andhra Pradesh State Government Scholarship

- Prestigious state-funded merit scholarship covering full tuition for the M.S. at NYU.

NYU Tandon Merit Scholarship

- Awarded at admission for academic excellence and research contributions in robotics and AI.

Smart India Hackathon (Hardware Edition): [National 1st Prize](#)

- National winner among 70,000+ teams for developing a flood-rescue drone system later adopted by the Uttarakhand State Disaster Response Force; awarded a INR 100,000 cash prize.

Titan Wear Hacks: National Top 10 Innovation Award

- Developed a body-heat-powered watch that detects assault & accidents and sends automated emergency alerts.

India International Science Festival (Student Engineering Model Competition) - 2nd Place

- For a long-range drone system enabling up to 20 km control and telemetry via LoRa-based communication.

Organizer - [EMAI](#) Workshop, IEEE International Conference on Image Processing (ICIP) 2024

- Co-organized a peer-reviewed international workshop on Embodied AI at the IEEE ICIP

MA'AN Social Incubator Selection

- Led the proposal for ChatSign, a real-time sign-language translation system, selected for competitive incubator funding and development.

NCC Army Wing – Best Cadet Award

- Recognized among top national cadets for excellence in drill, firing, written tests, leadership, and interviews.

National Handball Player

- Played at the national level, representing my state in several major handball tournaments.